



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Main)

Full Syllabus

19-07-2021

JEE(Main + Advanced) : Leader & Enthusiast Course (PHASE : SCORE-3.0)

Test Type : SCORE-3.0

ANSWER KEY

PHYSICS

	Q.	1	2	3	4	5	6	7	8	9	10
SEC-1	A.	B	A	C	C	D	A	B	C	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	A	B	C	A	C	D	D	B	D
SEC-2	Q.	1	2	3	4	5	6	7	8	9	10
	A.	7.00	7.00	50.00	8.00	3.00	5.00	5.00	2.00	3.00	4.00

CHEMISTRY

	Q.	1	2	3	4	5	6	7	8	9	10
SEC-1	A.	C	C	C	A	B	D	B	D	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	C	C	D	D	D	C	D	D
SEC-2	Q.	1	2	3	4	5	6	7	8	9	10
	A.	9.36	94.00	4.00	8.00	4.08	74.50	3.15	4.00	400.00	3.00

MATHEMATICS

	Q.	1	2	3	4	5	6	7	8	9	10
SEC-1	A.	D	A	A	D	B	D	A	C	A	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	C	B	C	B	B	A	C	C	C
SEC-2	Q.	1	2	3	4	5	6	7	8	9	10
	A.	0.71	2.50	0.50	5.00	4.00	0.09	2.00	1.00	16.00	0.24 or 0.25

HINT – SHEET

PHYSICS

SEC-1

1. Ans (B)

$$E = M \frac{di}{dt}, M = \text{mutual inductance}$$

$$\frac{E}{50 \times 10^{-3}} = \frac{6/0.02}{8/0.5}$$

$$E = 0.94V$$

2. Ans (A)

$$\lambda = \frac{\ln 2}{12.7h} = 5.46 \times 10^{-2} h^{-1}$$

$$\lambda = \lambda_{EC} + \lambda_{\beta^-} = \lambda_{EC} + \frac{39}{61} \lambda_{EC}$$

$$\lambda_{EC} = 3.329 \times 10^{-2} h^{-1}$$

$$t_{\text{half EC}} = \frac{\ln 2}{\lambda_{EC}} = 20.8h$$

$$\lambda_{\beta^-} = \frac{39}{61} \lambda_{EC} = 2.128 \times 10^{-2} h^{-1}$$

$$t_{\text{half EC}} = \frac{\ln 2}{\lambda_{\beta^-}} = 32.6h$$

3. **Ans (C)**

$$f = 25 \text{ cm}$$

$$m = \frac{-v}{u} = \frac{f}{f - u}$$

$$m = \frac{25}{25 - (-25)} = 0.5$$

4. **Ans (C)**

$$t = \sum \frac{\ell_i}{\sqrt{\frac{F}{\mu}}}$$

$$t = \sum \frac{\ell_i}{\sqrt{F}} \sqrt{\frac{m_i}{\ell_i}}$$

$$t = \sum \frac{\sqrt{m_i \ell_i}}{\sqrt{F}}$$

$$t = \sqrt{\frac{m\ell}{F}} \left[1 + \frac{1}{2} + \frac{1}{4} + \dots \right] = 2\sqrt{\frac{m\ell}{F}}$$

5. **Ans (D)**

frequencies are $f, f + 2, f + 4, \dots, f + 100(2)$

$$f + 200 = 3f$$

$$f = 100 \text{ Hz}$$

$$\text{Tenth fork has frequency} = f + 2(9) = 118 \text{ Hz}$$

6. **Ans (A)**

force balance for liquid column above upper

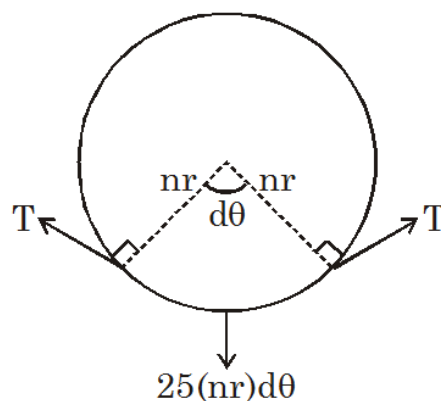
half of sphere

$$P_0 \pi r^2 + \rho V g = F$$

$$P_0 \pi r^2 + \rho \left[\pi r^2 (3r) - \frac{2\pi r^3}{3} \right] g = F$$

$$F = \frac{\pi r^2}{3} [3P_0 + 7\rho g]$$

7. **Ans (B)**



force balance of arc element

$$2T \sin \left(\frac{d\theta}{2} \right) = 2S n r d\theta$$

$$T = 2S n r \text{ (tension in band)}$$

$$T = k(\text{elongation})$$

$$2S n r = k[2\pi n r - 2\pi r]$$

$$k = \frac{S n}{\pi(n-1)} \text{ (elastic force constant of band)}$$

force balance of mass m

$$kx = mg$$

$$\frac{S n}{\pi(n-1)} (2\pi r) = mg$$

$$S = \frac{mg(n-1)}{2nr}$$

8. **Ans (C)**

torque by wire = magnetic torque

$$k\delta = mB_h \sin \alpha$$

k = torsion constant, δ = net twist

m = magnetic moment,

B_h = horizontal component of Earth's magnetic field

α = angle between magnetic moment and magnetic meridian

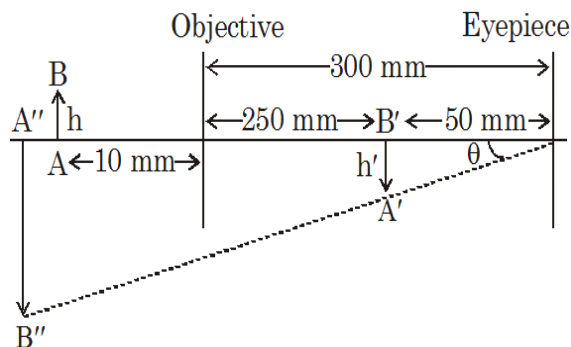
$$\frac{m_1}{m_2} = \frac{\delta_1}{\delta_2} = \frac{180 - 30}{270 - 30} = 5/8$$

9. **Ans (B)**

$$I = \frac{P}{4\pi r^2} = \frac{E_{rms}^2}{c\mu_0}$$

$$E_{rms} = \sqrt{\frac{250 \times 3 \times 10^8 \times 4\pi \times 10^{-7}}{4\pi(0.9)^2}} = 96V/m$$

10. **Ans (A)**



$$\frac{h'}{h} = \frac{250}{10}$$

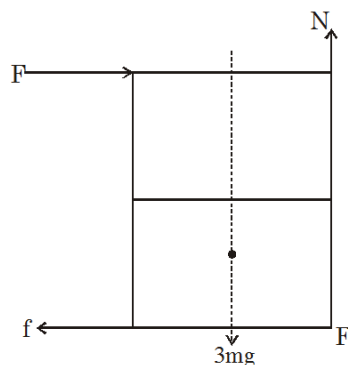
$$h' = 25h$$

$$\theta = \frac{h'}{50} = \frac{25h}{50} = \frac{h}{2}$$

$$M = \frac{\theta}{\theta_0} = \frac{h/2}{h/250} = 125X$$

12. **Ans (A)**

Equilibrium equations for toppling before sliding.
Normal reaction passes through F



$$F = f$$

$$N = 3mg$$

$$\tau_F = 0$$

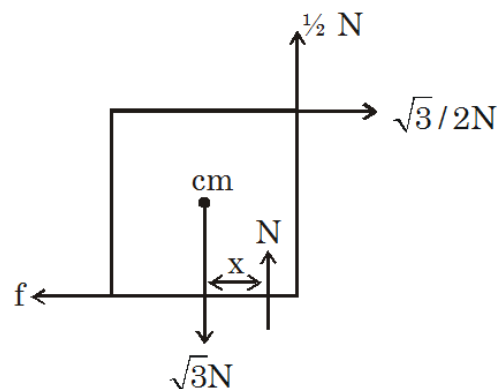
$$f(2h) = 3mg \left(\frac{h}{2} \right)$$

$$\text{but } f \leq \mu N$$

$$\frac{3mg}{4} \leq \mu (3mg)$$

$$\mu \geq 0.25$$

13. **Ans (B)**



$$N = \sqrt{3} - 1/2 \quad \dots(1)$$

$$f = \sqrt{3}/2 \quad \dots(2)$$

$$\tau_{cm} = 0$$

$$f(10) + \frac{\sqrt{3}}{2} \times 10 = \frac{1}{2} \times 10 + Nx$$

$$x = 10 \text{ cm}$$

14. **Ans (C)**

Net electric field should be normal to conductor's surface

Net electric field due to all the outside charges is zero inside a conducting shell

15. **Ans (A)**

$$y_{cm} = \frac{\int_0^\pi \lambda_0 \sin \theta R d\theta \sin \theta}{\int_0^\pi \lambda_0 \sin \theta R d\theta} = \frac{R\pi}{4}$$

$$x_{cm} = 0 \text{ by symmetry.}$$

16. **Ans (C)**

$$mg = B + FV$$

$$mg = \frac{mg}{2} + FV$$

$$\text{W.d. by } mg = mgx$$

$$\text{W.d. by } FV = -\frac{mg}{2}$$

$$\text{ratio} = \frac{-1}{2}$$

17. Ans (D)

kinetic energy of electron

$$K = \frac{Qe}{4\pi\epsilon_0 r}$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}}$$

$$\lambda = \frac{h}{\sqrt{2m \left(\frac{Qe}{4\pi\epsilon_0 r} \right)}}$$

$$Q = 100 \mu C$$

18. Ans (D)

upper lens

$$\frac{1}{f_1} = (1.5 - 1) \left(\frac{2}{R} \right) \text{ [Lens makers formula]}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{-4R} = \frac{1}{R} \Rightarrow v = 4R/3$$

$$\frac{1}{f_2} = (1.8 - 1) \left(\frac{2}{R} \right) \text{ [Lens makers formula]}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{-4R} = \frac{1.6}{R} \Rightarrow v = 20R/27$$

distance between images is

$$\frac{4R}{3} - \frac{20}{27}R = \frac{16R}{27}$$

19. Ans (B)

Initially

$$n_h = n_c = 7 \times 10^{15} \text{ m}^{-3}$$

$$\text{total charge carries} = 14 \times 10^{15} \text{ m}^{-3}$$

$$\text{after doping, donor impurity} = \frac{5 \times 10^{28}}{10^7} \text{ m}^{-3}$$

$$\text{number of electrons} = \frac{5 \times 10^{28}}{2 \times 10^7} \text{ m}^{-3}$$

$$n_c \gg n_h$$

$$\text{total charge carries} = n_c$$

$$= 2.5 \times 10^{21} \text{ m}^{-3}$$

fractional change

$$= \frac{2.5 \times 10^{21} - 14 \times 10^{15}}{14 \times 10^{15}} = 1.8 \times 10^5$$

20. Ans (D)

$$\text{Attenuation per km} = \frac{10 \log_{10} \left(\frac{P_{in}}{P_{out}} \right)}{\ell}$$

$$= \frac{10 \log_{10} \left(\frac{200}{10} \right)}{5} = 2.6 \text{ dB/km}$$

PHYSICS

SEC-2

1. Ans (7.00)

p.d. across 3 ohm or 1 microfarad

$$= 9 \left(\frac{3}{3+6} \right) = 3V$$

Its charge = 3 micro coulomb

p.d. across 7 ohm or 3 microfarad

$$= 9 \left(\frac{7}{7+2} \right) = 7V$$

Its charge = 21 μC

2. Ans (7.00)

Around Ampere's loop of radius 3a

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

$$B(2\pi(3a)) = \mu_0 I - \mu_0 I \left[\frac{\pi(3a)^2 - \pi(2a)^2}{\pi(4a)^2 - \pi(2a)^2} \right]$$

$$B = \frac{7\mu_0 I}{72\pi a}$$

3. **Ans (50.00)**

violet has $\mu = 1.8$

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = \frac{1}{v} - \frac{1}{u}$$

$$(1.8 - 1) \left(\frac{1}{40} - \frac{1}{-40} \right) = \frac{1}{v} - \frac{1}{-50}$$

$v = 50 \text{ cm}$

4. **Ans (8.00)**

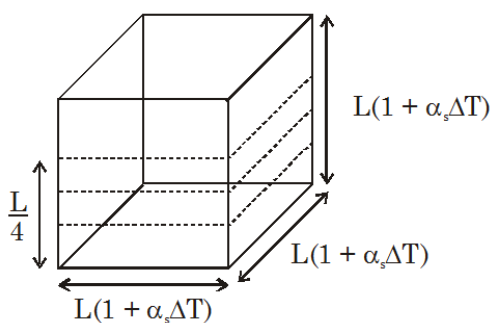
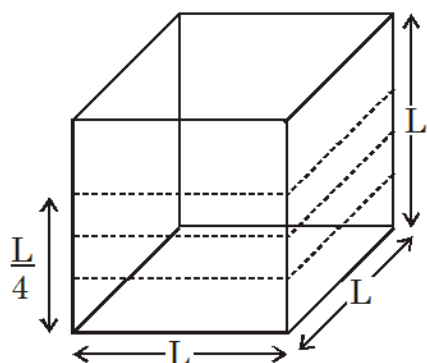
optical path difference

$$= (\mu_{\text{glass}} - 1)t_{\text{glass}} + (\mu_{\text{water}} - 1)t_{\text{water}} + (\mu_{\text{glass}} - 1)t_{\text{glass}}$$

$$= (1.5 - 1)(1) + (4/3 - 1)(21) + (1.5 - 1)(1)$$

$$= 8 \text{ cm}$$

5. **Ans (3.00)**



$$\frac{L^3}{4} (1 + \gamma_l \Delta T) = [L (1 + \alpha_s \Delta T)]^2 \left[\frac{L}{4} \right]$$

$$(1 + \gamma_l \Delta T) = [(1 + 2\alpha_s \Delta T)]$$

$$\alpha_s = \gamma_l / 2$$

6. **Ans (5.00)**

$$\text{least count} = \frac{\text{pitch}}{\text{number of CSD}} = \frac{0.01 \text{ m}}{2000}$$

$$= 5 \mu \text{ m}$$

7. **Ans (5.00)**

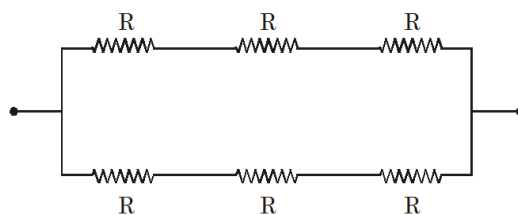
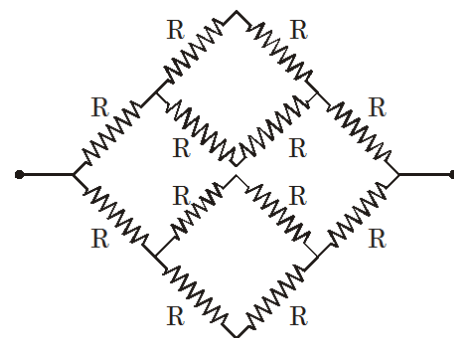
Flux through bottom circular end

$$\frac{q}{4\pi\epsilon_0} (2\pi(1 - \cos 37^\circ)) = \frac{q}{10\epsilon_0}$$

Flux through curved surface is

$$\frac{q}{2\epsilon_0} - \frac{q}{10\epsilon_0} = \frac{2q}{5\epsilon_0}$$

8. **Ans (2.00)**



For maximum power

$$R_{\text{eq}} = r_{\text{int}}$$

$$\frac{3R}{2} = 3$$

$$R = 2\Omega$$

9. **Ans (3.00)**

$$\text{energy} = \frac{kA^2}{2}$$

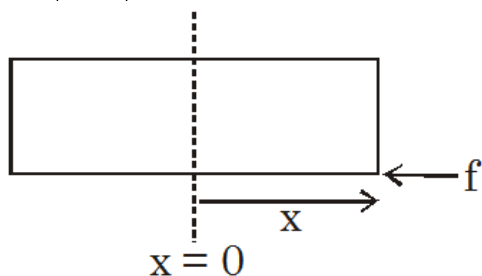
$$\frac{(k/4) A^2}{2} = \frac{k_{\text{eq}} A^2}{2}$$

$$\frac{1}{k_{\text{eq}}} = \frac{4}{k}$$

$$\frac{1}{k'} + \frac{1}{k} = \frac{4}{k}$$

$$k' = k/3$$

10. Ans (4.00)



At any instant frictional force is

$$f = \frac{\mu mgx}{L}$$

Work energy theorem

$$FL - \int_0^L \frac{\mu mgx}{L} dx = \frac{1}{2}mv^2$$

$$v = \sqrt{\mu gL}$$

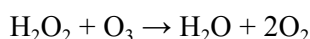
CHEMISTRY

SEC-1

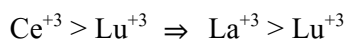
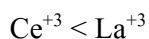
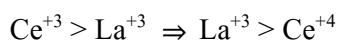
5. Ans (B)

In $\ddot{N}O_2$ & $\ddot{Cl}O_3$, valence e^- also form π -bond.

15. Ans (D)



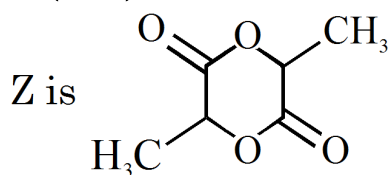
20. Ans (D)



CHEMISTRY

SEC-2

4. Ans (8.00)



10. Ans (3.00)

Acidic hydrogen liberate methane from CH_3 Mg Br.

MATHEMATICS

SEC-1

1. Ans (D)

$$-[x]^2 + 5[x] - 6 \geq 0$$

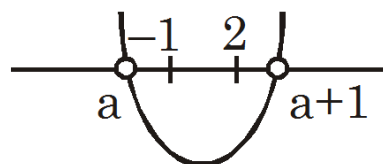
$$\Rightarrow [x]^2 - 5[x] + 6 \leq 0 \Rightarrow 2 \leq [x] \leq 3$$

$$\Rightarrow 2 \leq x < 4$$

and $\sin x \geq 0$ when $x \in [2n\pi, (2n+1)\pi]$

$$\therefore x \in [2, \pi]$$

2. Ans (A)



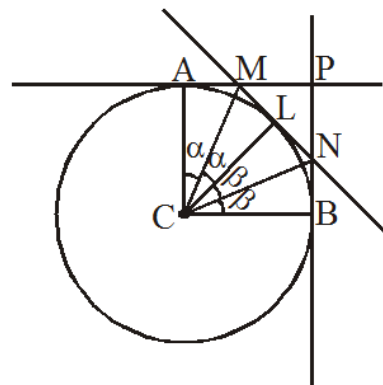
$$(x-a)(x-(a+1)) < 0 \quad \forall x \in (-1, 2)$$

$$\Rightarrow a \leq -1 \text{ and } a+1 \geq 2$$

$$\Rightarrow a \leq -1 \text{ and } a \geq 1$$

$$\Rightarrow a \in \phi$$

3. Ans (A)



$$\text{Since, } 2\alpha + 2\beta = 90^\circ$$

$$\Rightarrow a + b = 45^\circ$$

$$\Rightarrow \angle MCN = 45^\circ$$

4. Ans (D)

$$\vec{a} = \hat{i} + 2\hat{j} + (3-\pi)\hat{k}$$

$$\text{and } \vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$$

$$\text{So, } \vec{a} \cdot \vec{b} = 1 + 4 + 9 - 3\pi = 14 - 3\pi$$

5. Ans (B)

$$(1 + {}^7C_1x^4 + \dots)(1 + x^2)^3(1-x)^3$$

$$(1 + {}^7C_1x^4)(1 + x^6 + 3x^4 + 3x^2)(1 - x^3 + 3x^2 - 3x)$$

$$\therefore \text{coefficient of } x^5$$

$$= -9 - 3 - 21 = -33$$

6. **Ans (D)**

$$\sim (\sim p \vee (\sim q \vee r)) \\ = p \wedge (\sim (\sim q \vee r)) = p \wedge (q \wedge \sim r)$$

7. **Ans (A)**

$$e^2 = 1 + \sqrt{\sin^4 \alpha + \cos^4 \alpha} \\ = 1 + \sqrt{1 - 2\sin^2 \alpha \cdot \cos^2 \alpha} \\ = 1 + \sqrt{1 - \frac{1}{2} \cdot \sin^2 2\alpha}$$

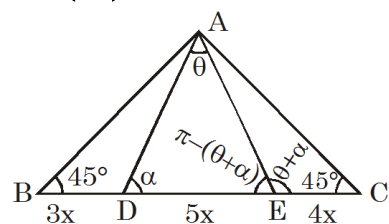
is maximum when $\sin 2\alpha = 0$

$$\therefore b^2 = 1 \quad \therefore 2b = 2$$

8. **Ans (C)**

$$\frac{x^2 + 2x^3 + \frac{1}{x} + \frac{1}{x^4}}{4} \geq \sqrt[4]{x^2 \cdot 2x^3 \cdot \frac{1}{x} \cdot \frac{1}{x^4}} = \sqrt[4]{2} \\ \Rightarrow x^2 + 2x^3 + \frac{1}{x} + \frac{1}{x^4} \geq 4\sqrt[4]{2} > 4$$

9. **Ans (A)**



In $\triangle ABE$:

$$(3 + 5)\cot \alpha = 5 + 3\cot(\theta + \alpha) \dots (i)$$

In $\triangle ADC$:

$$9\cot(\theta + \alpha) = 4\cot \alpha - 5$$

$$\Rightarrow 3[8\cot \alpha - 5] = 4\cot \alpha - 5 \quad (\text{from (i)})$$

$$\Rightarrow \cot \alpha = \frac{1}{2}$$

$$\therefore \tan(\theta + \alpha) = -3$$

$$\frac{\tan \theta + 2}{1 - 2 \tan \theta} = -3$$

$$\Rightarrow \tan \theta + 2 = -3 + 6 \tan \theta \Rightarrow \tan \theta = 1 \Rightarrow \theta = 45^\circ$$

10. **Ans (A)**

$$\begin{vmatrix} 2 & 1 & 0 \\ k & 2k & k-1 \\ k+1 & k & 1 \end{vmatrix} = 0$$

$$\Rightarrow 2(2k - k^2 + k) - 1(k - k^2 + 1) = 0$$

$$\Rightarrow 4k - 2k^2 + 2k - k + k^2 - 1 = 0$$

$$\Rightarrow -k^2 + 5k - 1 = 0$$

$$\Rightarrow k^2 - 5k + 1 = 0$$

$$\Rightarrow k = \frac{5 \pm \sqrt{25-4}}{2} = \frac{5 \pm \sqrt{21}}{2}$$

11. **Ans (B)**

Let x_i be the actual (correct) measurement

$$\text{So, given that } \frac{\sum (x_i + 3)}{10} = 20$$

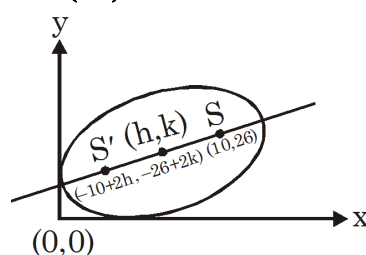
$$\text{and } \frac{\sum (x_i + 3)^2}{10} - (20)^2 = 15$$

$$\text{Hence, } \frac{\sum x_i}{10} = 17$$

$$\text{and, } \frac{\sum x_i^2}{10} - (17)^2 = 15$$

(note that variance is independent of shifting of origin)

12. **Ans (C)**



$$(-26 + 2k)(26) = (-10 + 2h)(10)$$

$$\Rightarrow (k - 13)13 = (h - 5)5$$

$$\Rightarrow 5h - 13k + 169 - 25 = 0$$

$$\Rightarrow 5x - 13y + 144 = 0$$

13. **Ans (B)**

$$\frac{dy}{dx} = xy^2 + x - y^2 \sin x - \sin x$$

$$\Rightarrow \frac{dy}{dx} = (x - \sin x)(y^2 + 1)$$

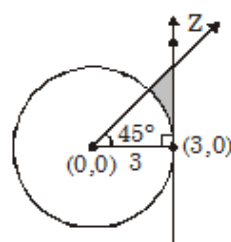
$$\text{Hence, } \int \frac{dy}{y^2 + 1} = \int (x - \sin x) dx$$

$$\Rightarrow \tan^{-1} y = \frac{x^2}{2} + \cos x + C$$

$$\text{Since, } y(0) = 0 \Rightarrow C = -1$$

$$\text{So, } y = \tan \left(\frac{x^2}{2} + \cos x - 1 \right)$$

14. **Ans (C)**



$$\text{shaded area} = \frac{1}{2} \times 3 \times 3 - \frac{\pi \times 3^2}{8} = \frac{36 - 9\pi}{8}$$

15. **Ans (B)**

$$f(0) = \lim_{x \rightarrow 0} f(x) = 0$$

$$\text{Also, } f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = 0$$

16. **Ans (B)**

$$x_1 + x_2 + x_3 + x_4 + x_5 = 7$$

So, required number of ways

$$= {}^{7+5-1}C_{5-1} = {}^{11}C_4 = {}^{11}C_7$$

17. **Ans (A)**

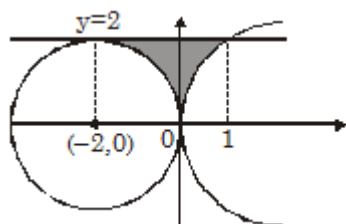
$$\cos x \cdot \sin x \cdot \cos 2x = \frac{\sqrt{2}}{8} = \frac{1}{4\sqrt{2}}$$

$$\sin 2x \cdot \cos 2x = \frac{1}{2\sqrt{2}} \Rightarrow \sin 4x = \frac{1}{\sqrt{2}}$$

$$\therefore 4x = n\pi + (-1)^n \frac{\pi}{4}$$

$$\Rightarrow x = \frac{n\pi}{4} + (-1)^n \frac{\pi}{16}, n \in \mathbb{I}$$

18. **Ans (C)**



$y = 2$ and $y^2 = 4x$ meet at $(1, 2)$

$$A = \left(4 - \frac{\pi(2)^2}{4} \right) + \int_0^2 \frac{y^2}{4} dy$$

$$= 4 - \pi + \frac{1}{4.3} \cdot 2^3 = 4 - \pi + \frac{2}{3} = \frac{14}{3} - \pi$$

19. **Ans (C)**

Let the common tangent be $y = mx + \frac{12}{m}$

$$\text{So, } \left(\frac{12}{m} \right)^2 = 72(1 + m^2)$$

$$\Rightarrow m^4 + m^2 - 2 = 0$$

$$\Rightarrow m^2 = 1 \Rightarrow \frac{12}{m} = \pm 12$$

20. **Ans (C)**

$$f'(x) = 4\cos x - 4\cos x \sin x$$

$$= 4\cos x (1 - \sin x)$$

$$\therefore \text{It is non-monotonic in } \left(\frac{\pi}{2}, \frac{5\pi}{3} \right)$$

MATHEMATICS

SEC-2

1. **Ans (0.71)**

$$P = 1 - \left[\frac{2}{10} \times \frac{8}{9} \times \frac{1}{8} + \frac{4}{10} \times \left[\frac{3}{9} \times \frac{2}{8} + \frac{6}{9} \times \frac{3}{8} \right] \times 2 \right]$$

$$= 1 - \left(\frac{16 + 48 + 144}{10.9.8} \right) = 1 - \frac{208}{10.9.8}$$

$$= 1 - \frac{26}{10.9} = 1 - \frac{13}{45} = \frac{32}{45}$$

2. **Ans (2.50)**

$$\lim_{x \rightarrow 0} \frac{\frac{(2p-3)}{x} + \frac{\sin x}{x}}{2 - \frac{\tan x}{x}} = q$$

$$\therefore p = \frac{3}{2} \text{ and } q = \frac{1}{2-1} = 1$$

$$\text{So, } p + q = \frac{5}{2}$$

3. **Ans (0.50)**

Differentiating w.r.t x

$$f(x) \cdot [\sin^2 x + \cos x] = 2 f(x) \cdot f'(x)$$

$$\therefore f'(x) = \frac{\sin^2 x + \cos x}{2}, f(x) \neq 0$$

$$\lim_{x \rightarrow 0} f'(x) = \frac{1}{2} = f'(0) \text{ (as } f'(x) \text{ is continuous)}$$

Also, note that $f(0) = 0$

4. **Ans (5.00)**

$$\{1, 3\} \times \{2, 4\} \subseteq A \times B$$

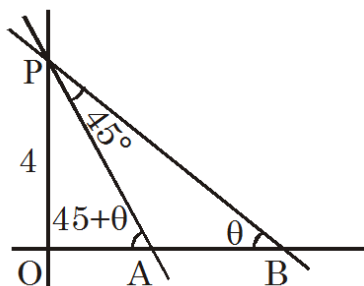
$$\Rightarrow \{(1, 2), (1, 4), (3, 2), (3, 4)\} \subseteq \{(1, 3), (1, 4),$$

$$(1, y), (2, 3), (2, 4), (2, y), (x, 3), (x, 4), (x, y)\}$$

$$\Rightarrow y = 2, x = 3$$

5. Ans (4.00)

$$AB = 4\cot\theta - 4\cot(45^\circ + \theta)$$



$$= 4 \left[\cot\theta - \frac{\cot\theta - 1}{\cot\theta + 1} \right]$$

$$= 4 \left[(\cot\theta + 1) - 1 - \left(\frac{(\cot\theta + 1) - 2}{\cot\theta + 1} \right) \right]$$

$$\Rightarrow AB = 4 \left[(\cot\theta + 1) + \frac{2}{(\cot\theta + 1)} - 2 \right]$$

$$\text{Now, } 0 < \theta < \frac{\pi}{2} \text{ and } 0 < \theta + \frac{\pi}{4} < \frac{\pi}{2}$$

$$\Rightarrow 0 < \theta < \frac{\pi}{4}$$

$$\text{So, } AB = f(x) = 4 \left[x + \frac{2}{x} - 2 \right]$$

$$x = \cot\theta + 1 \in (2, \infty)$$

$$f'(x) = 4 \left[1 - \frac{2}{x^2} \right] > 0$$

$\therefore f(x)$ is increasing function

$$f(x) > f(2) = 4$$

6. Ans (0.09)

$$\int_0^1 (1-2x)^{10} dx$$

$$= \int_0^1 [{}^{10}C_0 - {}^{10}C_1(2x) + {}^{10}C_2(2x)^2 - {}^{10}C_3(2x)^3$$

$$+ \dots + {}^{10}C_{10}(2x)^{10}] dx$$

$$\Rightarrow \left[\frac{(1-2x)^{11}}{-22} \right]_0^1 = {}^{10}C_0 - {}^{10}C_1 + \frac{{}^{10}C_2 \cdot 2^2}{3} - \frac{{}^{10}C_3 \cdot 2^3}{4}$$

$$+ \dots + \frac{{}^{10}C_{10} 2^{10}}{11}$$

7. Ans (2.00)

$$\frac{a+1}{-2a-3} = \frac{2}{a} \neq \frac{a+2}{a^2-1}$$

$$\Rightarrow a^2 + a = -4a - 6$$

$$\Rightarrow a^2 + 5a + 6 = 0 \Rightarrow (a+2)(a+3) = 0$$

$$\Rightarrow a = -2 \text{ or } -3$$

8. Ans (1.00)

$$2\sin^{-1}\sqrt{x} = \cos^{-1}x^2$$

$$\Rightarrow \cos(2\sin^{-1}\sqrt{x}) = \cos(\cos^{-1}x^2)$$

$$\Rightarrow 1 - 2(\sqrt{x})^2 = x^2 \quad (0 \leq x \leq 1)$$

$$\Rightarrow 1 - 2x = x^2$$

$$\Rightarrow x^2 + 2x - 1 = 0$$

$$\Rightarrow x = \frac{-2 \pm \sqrt{8}}{2} = -1 \pm \sqrt{2}$$

$$\therefore x = \sqrt{2} - 1$$

9. Ans (16.00)

$$|M| = -2ab - 1(-ab - 2ab)$$

$$= -2ab + ab + 2ab = ab$$

$$\text{So, } \det(\text{Adj.}M) = |M|^2 = (ab)^2$$

$$\therefore \text{Its maximum value is } (2 \cdot 2)^2 = 16$$

10. Ans (0.24 or 0.25)

$$\sin A = \frac{3}{5} \Rightarrow \cos A = \pm \frac{4}{5}$$

$$\text{Also, } \cos B = \frac{5}{13} \Rightarrow \sin B = \frac{12}{13}$$

$$\text{Now, } \sin C = \sin(A+B)$$

$$= \sin A \cos B + \cos A \sin B$$

$$= \frac{3}{5} \cdot \frac{5}{13} \pm \frac{4}{5} \cdot \frac{12}{13}$$

$$= \frac{63}{65} \left(-\frac{23}{65} \text{ is not possible} \right)$$

$$\Rightarrow \cos A = \frac{4}{5}$$

$$\therefore \cos C = -[\cos A \cos B - \sin A \sin B]$$

$$= -\frac{4}{5} \cdot \frac{5}{13} + \frac{3}{5} \cdot \frac{12}{13} = \frac{16}{65}$$